

Concurrent Aerobic and Strength Training

Scientific Basics
and Practical Applications

Moritz Schumann
Bent R. Rønnestad
Editors



Springer

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Moritz Schumann
Department of Molecular and Cellular
Sports Medicine
German Sport University
Cologne
Germany

Bent R. Rønnestad
Department of Sports Sciences
Lillehammer, Inland Norway University of
Applied Sciences
Lillehammer
Norway

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Preface

Endurance and strength create the foundation of sports performance and are considered as basic elements of physical fitness and health. In addition, global exercise recommendations strongly recommend both aerobic and strength training for improvements of physical fitness and health as well as the prevention of chronic diseases across the life span. However, aerobic and strength training induce dissimilar biological adaptations, which may in combination result in compromised training adaptations. Thus, designing concurrent aerobic and strength training programs for various populations requires special consideration.

Typically, exercise and health professionals as well as sport practitioners are similarly concerned about the optimal concurrent training mode in an attempt to maximize both aerobic and muscular adaptations. Most frequently, questions such as whether concurrent aerobic and strength training should be performed on the same day or separated onto different days or whether endurance training performed first in a training session may negatively affect the quality of a subsequent strength training session and, thus, compromises long-term adaptations are controversially discussed. Furthermore, the importance of strength training for athletic performance of both team and individual sports has not yet received thorough acceptance among coaches and athletes.

While concurrent training has been practically applied in exercise training and sports coaching for multiple decades, it has been of scientific interest for a much shorter time. In fact, it was not until 1980 when Robert C Hickson found that strength but not endurance development may be compromised when intensive running and strength training were performed concurrently [1]. Ever since his pioneering study this phenomenon is known as the “interference effect” and has been of extensive scientific interest.

Interestingly, in contrast to these early observations, more recent studies have actually shown that concurrent training may not necessarily compromise neuromuscular adaptations to the extent that was initially suggested. Much rather, it was shown that the magnitude of blunted neural and muscular adaptations appears to be depending on a complex interplay of training program variables and biological predispositions. This was also underlined by scientific studies in which strength training was shown to be beneficial for endurance athletes by enhancing aerobic performance.

Nearly four decades after the appearance of the initial studies on concurrent aerobic and strength training, a large body of scientific studies on this topic is available. Nevertheless, numerous myths and false assumptions exist in fitness and health communities but also among top-level sports coaches and athletes, much of which is fueled by non-evidence-based knowledge. *Concurrent Aerobic and Strength Training: Scientific Basics and Practical Applications* aims to elucidate the concept of concurrent training for physical fitness and sports performance by providing a comprehensive overview on the latest stand of research.

Divided into 5 parts and 27 chapters, this book provides an extensive guide for exercise and health professionals, students, scientists, sport coaches, athletes of various sports, and those with a general interest in concurrent aerobic and strength training. Following a brief historical overview of the past decades of research on concurrent training, in Part I the physiological and neuromuscular differences of endurance and strength training are discussed. Thereafter, Part II aims at providing an up-to-date analysis of existing explanations for the interference phenomenon, while in Part III the training-methodological difficulties of combined aerobic and strength training are elucidated. In Parts IV and V the theoretical considerations reviewed in the previous parts are then practically applied to specific populations, ranging from children and elderly to athletes of various sports.

With *Concurrent Aerobic and Strength Training: Scientific Basics and Practical Applications*, we were privileged to work together with leading scientists and coaches from across the world to provide a novel book on one of the “hot topics” of exercise training. Our highest priority was to make this book an easily understandable and at the same time scientifically supported guide for the daily practice. We sincerely hope you will share our joy when reading this book.

Cologne, Germany
Lillehammer, Norway

Moritz Schumann
Bent R. Rønnestad

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